is also important to remember that, even for normalised vectors, $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same state. The global phase $e^{i\theta}$ has no physical significance.

Because $\mathcal{H}$ is a complex vector space, for any pair of states $|\psi\rangle$ and $|\phi\rangle$, any linear combination $\alpha|\psi\rangle + \beta|\phi\rangle$ with $\alpha, \beta \in \mathcal{C}$ is also a state. This is known as the *superposition principle*. Note that, as opposed to a global phase, the relative phase in such a superposition *is* physically significant. That is, $\alpha|\psi\rangle + \beta|\phi\rangle$ and $\alpha|\psi\rangle + e^{i\theta}\beta|\phi\rangle$ are not the same state.

An orthornomal basis for $\mathcal{H}$ is a discrete set of vectors $\{|n\rangle\}_n$ with $\langle n|m\rangle = \delta_{nm}$ where δ_{nm} is the Kroenecker delta function. Any state can then be expressed as

$$|\psi\rangle = \sum_n \alpha_n |n\rangle. \tag{2.1}$$

with $\alpha_n \in \mathcal{C}$. For normalised states, $\sum_n |\alpha_n|^2 = 1$. Vectors and their conjugates (kets and bras) can be represented by column and row vectors in a given basis

$$|\psi\rangle \sim \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \end{pmatrix} \qquad \text{and} \qquad \langle\psi| \sim (\alpha_0^*, \alpha_1^*, \cdots). \tag{2.2}$$

In finite dimension, the span of $\{|n\rangle\}_n$ equals $\mathcal{H}$ and it is thus a basis in the usual sense of linear algebra. In infinite dimensions, the span of $\{|n\rangle\}_n$ is required to be dense in $\mathcal{H}$ (completeness). It is also sometimes possible, and useful, to instead express states in terms of a continuous set of vectors (although dealing with normalisation becomes more subtle).

2.1.2 Evolution and operators

Recall that an operator $\hat{A}$ on $\mathcal{H}$ is a linear map from vectors to vectors

$$\hat{A}(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha\hat{A}|\psi\rangle + \beta\hat{A}|\phi\rangle. \tag{2.3}$$

In these notes, we use symbols with hats ($\hat{\cdot}$) to indicate operators. The Hermitian conjugate (or adjoint) $\hat{A}^\dagger$ of an operator $\hat{A}$ is defined by

$$\langle\hat{A}^\dagger\phi|\psi\rangle = \langle\phi|\hat{A}\psi\rangle \tag{2.4}$$

for all vectors $|\psi\rangle$ and $|\phi\rangle$, where $|\hat{A}\psi\rangle$ denotes $\hat{A}|\psi\rangle$ and similarly for conjugate vectors. An operator is Hermitian (or self-adjoint) if $\hat{A}^\dagger = \hat{A}$ and it is unitary if $\hat{A}^\dagger\hat{A} = \hat{A}\hat{A}^\dagger = \mathbb{1}$, where $\mathbb{1}$ is the identity operator. An operator is positive semidefinite if $\langle\psi|\hat{A}|\psi\rangle \geq 0$ for all vectors $|\psi\rangle$. In a given orthonormal basis $\{|n\rangle\}_n$, any operator $\hat{A}$ can be represented by a matrix with elements $A_{mn} = \langle m|\hat{A}|n\rangle$. The Hermitian conjugate operator has matrix elements $(\hat{A}^\dagger)_{mn} = A^*_{nm}$, i.e. it corresponds to the transpose, conjugate matrix.

The time evolution of a closed quantum system is generated by a Hermitian operator $\hat{H}$ known as the Hamiltonian and is governed by the time-dependent Schrödinger equation

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle, \tag{2.5}$$

where $\hbar$ is the reduced Planck constant. It follows that evolution in closed quantum systems is always unitary, i.e.

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle \tag{2.6}$$

for some family of unitary operators $\hat{U}(t)$. If the Hamiltonian is independent of time then $\hat{U}(t) = e^{-i\hat{H}t}$ (in units where $\hbar = 1$).

Because of the unitarity of time evolution, all the basic operations in quantum computing and information processing (except measurements) are described by unitary operators.

2.1.3 Measurements and observables

The *measurement postulate* dictates how measurements on physical systems are modelled in quantum mechanics. We can describe measurements in two seemingly different but ultimately equivalent ways.

POVMs

A general measurement in quantum mechanics is associated with a set of measurement operators $\{\hat{M}_m\}$ where m labels the possible measurement outcomes. For a system in a normalised state $|\psi\rangle$, the probability for the outcome m to occur is

$$p(m) = \langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle. \tag{2.7}$$

This is known as the *Born rule*. Because this expression is the norm of the vector $\hat{M}_m|\psi\rangle$ (or equivalently, because the operator $\hat{E}_m = \hat{M}_m^\dagger \hat{M}_m$ is positive semidefinite), the probabilities are necessarily non-negative, $p(m) \geq 0$. In addition, the measurement operators must satisfy the completeness relation

$$\sum_m \hat{M}_m^\dagger \hat{M}_m = \mathbb{1}, \tag{2.8}$$

which ensures that the probabilities sum to one,

$$\sum_m p(m) = \sum_m \langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle = \langle\psi|(\sum_m \hat{M}_m^\dagger \hat{M}_m)|\psi\rangle = 1, \tag{2.9}$$

for any normalised state $|\psi\rangle$. The (normalised) state of the system immediately after a measurement with outcome m is given by

$$\frac{\hat{M}_m|\psi\rangle}{\sqrt{\langle\psi|\hat{M}_m^\dagger \hat{M}_m|\psi\rangle}} = \frac{1}{\sqrt{p(m)}}\hat{M}_m|\psi\rangle. \tag{2.10}$$

The set of positive operators $\{\hat{E}_m\}$ is called a *positive-operator-valued measure* (POVM). Any set of operators satisfying

$$\hat{E}_m^\dagger = \hat{E}_m \qquad \text{(Hermiticity)}, \tag{2.11}$$

$$\hat{E}_m \geq 0 \qquad \text{(positivity)}, \tag{2.12}$$

$$\sum_m \hat{E}_m = \mathbb{1} \qquad \text{(completeness)}, \tag{2.13}$$

form a POVM and the operators $\hat{E}_m$ are referred to as POVM elements. POVMs provide the most general description of measurements in quantum mechanics. The outcome probabilities are given by $p(m) = \langle\psi|\hat{E}_m|\psi\rangle$. Note that the decomposition of a POVM into measurement operators, $\hat{E}_m = \hat{M}_m^\dagger \hat{M}_m$,